

The Triadic Closure within Simulation Theory

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July 2026

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Abstract

Simulation theory traditionally assumes that simulated worlds must be organized around a single global spacetime, a universal causal structure, and a shared informational substrate. This paper challenges that assumption by demonstrating that any efficient, scalable, and observer-consistent simulation must instead implement *triadic closure*—the architectural mechanism underlying the Triadic Cycle of energy, geometry, and information. Earlier work within the Triadic Cosmos ecosystem established that physical reality is structured by horizon-bounded, triadically recursive closures whose emergent temporal flow arises from successive triadic updates. Here we show that modern simulation systems, including rendering engines, physics engines, and distributed computation frameworks, already instantiate the same architectural pattern: observer-centric geometry, local informational boundaries, energetic update constraints, and recursive triadic cycles.

We further demonstrate that simulations built according to triadic closure become self-similar fractals: each simulated observer generates its own closure, each closure supports its own triadic recursion, and simulations can contain further simulations without architectural contradiction. This yields a form of infinite simulational recursion that is structurally coherent, computationally optimal, and architecturally identical to the recursive triadic fractal structure of the physical universe. The convergence between physical reality and human-constructed simulations is therefore not coincidental: both domains adopt triadic closure because it is the minimal architecture capable of supporting consistent causal ordering, informational accessibility, energetic evolution, and emergent time.

By identifying triadic closure as the shared architectural foundation of both physical law and simulation design, this paper reframes simulation theory as a structural consequence of the triadic organization of reality. The universe is not merely simulatable; it is itself organized according to the same principles that make simulations possible. This perspective yields testable predictions, clarifies the relationship between physical and artificial worlds, and positions triadic closure as the minimal and universal grammar of both physics and simulation.

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1 Introduction

The search for a unified account of physical reality and artificial simulation has traditionally been fragmented across philosophy, physics, and computer science. Existing simulation frameworks—probabilistic arguments Bostrom 2003, algorithmic universe models Schmidhuber 2002, digital physics Fredkin 1990, and rendering-based theories Lanier 2017—lack a structural ontology capable of explaining how coherent worlds function, scale, and embed. None provide the architectural mechanisms required for emergent time, horizon-bounded observation, or nested simulations. As a result, simulation theory has remained conceptually provocative but architecturally incomplete.

This paper introduces *triadic closure* as the minimal and universal architectural grammar underlying coherent dynamical systems. Triadic closure identifies three irreducible structural roles—energy, geometry, and information—connected through recursive update. These roles form a closed cycle that generates emergent temporal flow, enforces locality, and bounds informational accessibility. We show that triadic closure is not merely sufficient for coherent worlds; it is *necessary*. Any system—physical or simulated—that supports scalable, observer-consistent dynamics must instantiate these roles.

A central contribution of this work is the demonstration that physical reality and artificial simulation are *structurally isomorphic*. Modern simulation engines (Unreal, Unity, idTech, Frostbite) already implement triadic closure implicitly: compute budget functions as energetic capacity, rendering frustums instantiate geometric horizons, scene graphs encode informational structure, and frame updates generate emergent time. This convergence is not accidental. It reflects the architectural necessity of triadic closure for any coherent world, regardless of substrate.

From this isomorphism we derive the *Simulation Indeterminacy Theorem*: no internal observer can determine whether their closure is fundamental or simulated. Because all coherent closures share the same triadic architecture and enforce the same horizon-bounded limits, simulation status becomes in principle undecidable and empirically unfalsifiable. This result reframes the nature of reality: fundamentality becomes closure-relative rather than absolute, and the distinction between “real” and “simulated” collapses into a single architectural class.

The architectural nature of triadic closure also yields a unified falsifiability framework. The Triadic Cosmos ontology is falsified if and only if a coherent dynamical system—physical or simulated—can be constructed that violates triadic closure. This provides simulation theorists with a direct falsification route: build a coherent simulation without triadic roles, without recursive update, or with global informational access. To date, no such system has ever been discovered. All known coherent simulations converge on triadic closure, providing empirical support for its universality.

2 Related Work

Simulation theory spans several distinct research traditions, ranging from philosophical arguments about simulated universes to technical work on rendering engines, distributed systems, and virtual environments. The present paper differs from these approaches by identifying *triadic closure* as the minimal architectural mechanism shared by both physical reality and artificial simulations. This section situates the proposal within the broader landscape of simulation theory, computational architectures, and the Triadic Cosmos ecosystem.

2.1 Triadic Cosmos Ecosystem

The present work builds directly on four core papers that established the architectural foundation of triadic closure. The Triadic Cycle was first introduced as a minimal ontology of energy,

geometry, and information, forming a closed interdependence $E \rightarrow G \rightarrow I \rightarrow E$ Witte 2026b. This foundational paper identified the triadic roles as irreducible structural constraints already implicit across modern physics.

The second paper extended this perspective by showing that the universe is organized as a recursive triadic fractal, with emergent time arising from successive triadic updates and with intelligence appearing as a fractal repetition of the same architecture Witte 2026c. This work established recursion as the generator of temporal structure and revealed the fractal organization of triadic closures across scales.

Local closure cosmology further developed the architectural framework by demonstrating that physical observation, causal propagation, recursion depth, and accessible information are inherently local rather than global Witte 2026a. This paper introduced horizon-bounded closures as the structural units of physical reality and explained why multiple independent scientific tracks find no new fundamental physics within the same observational domain.

The unification of general relativity and quantum mechanics through triadic closure was presented in the fourth paper Witte 2026d. There, GR and QM were shown to be complementary projections of the same triadic architecture, with local closure providing the structural mechanism through which geometric, energetic, and informational roles cohere. The present paper extends this architectural perspective into simulation theory, showing that artificial simulations constructed by intelligent systems naturally replicate the same triadic organization.

2.2 Simulation Theory and Nested Simulations

Classical simulation theory, including the well-known argument by Bostrom Bostrom 2003, typically assumes that simulated worlds are organized around global spacetime, universal causality, and shared informational substrates. Variants of the nested-simulation hypothesis (e.g., Tipler Tipler 2008, Schmidhuber Schmidhuber 2002) explore the possibility of recursively embedded simulations but do not provide an architectural mechanism capable of supporting infinite recursion without contradiction. These approaches lack a structural grammar linking informational, geometric, and energetic roles, and therefore cannot explain how nested simulations maintain causal coherence or emergent temporal structure.

The present work differs by showing that triadic closure provides exactly such a grammar. Simulations constructed according to triadic closure become recursive triadic fractals: each simulated observer generates its own closure, each closure supports its own triadic recursion, and simulations can contain further simulations without architectural inconsistency. This provides a minimal and falsifiable mechanism for infinite simulational recursion.

2.3 Rendering Engines and Observer-Centric Architectures

Modern rendering engines (Unreal Engine Games 2024, Unity Technologies 2024, idTech Carmack 2011) do not operate on global worlds. Instead, they implement observer-centric rendering, frustum culling, occlusion culling, and horizon-bounded geometry. Physics engines (Ode, Havok, Bullet) likewise simulate only the regions accessible to the observer, using locality, lazy evaluation, and horizon-bounded updates. These architectures instantiate the same structural principles as triadic closure: geometry as relational constraint, information as local accessibility, and energy as update capacity.

Distributed simulation frameworks (Fujimoto Fujimoto 2000, Jefferson Jefferson 1985) further demonstrate that global synchronization is computationally infeasible. Efficient simulation requires local causal domains, observer-specific state, and horizon-bounded consistency—precisely the structural features of triadic closure.

2.4 Virtual Worlds, VR Architectures, and Emergent Time

Virtual environments and VR systems (Lanier 2017, Slater 2020) rely on observer-dependent geometry, local informational boundaries, and energetic update constraints. Temporal flow in these systems emerges from recursive update cycles rather than from global time parameters, mirroring the emergent-time mechanism of triadic closure Witte 2026b; Witte 2026c. The architectural correspondence suggests that emergent time is not a contingent feature of artificial simulation but a structural consequence of triadic organization.

2.5 Structural and Computational Approaches

Structural realist and information-theoretic approaches (Ladyman and Ross 2007, Lloyd 2006) argue that informational and relational structure may be more fundamental than objects or fields. Computational models of physics (Fredkin 1990, Wolfram 2002) explore cellular automata and algorithmic universes but do not provide a closed architectural mechanism linking geometry, information, and energy. The Triadic Cycle fills this gap by identifying the minimal structural interdependence required for physical and simulational coherence.

2.6 Summary

Existing work on simulation theory, rendering engines, distributed systems, and virtual worlds provides partial realizations of observer-centric architecture, horizon-bounded geometry, and recursive update cycles. The Triadic Cosmos ecosystem provides the missing architectural grammar linking these features. The present paper integrates these strands by showing that triadic closure is the minimal and universal structure underlying both physical reality and artificial simulation, and that recursive triadic fractals arise naturally in both domains.

3 Triadic Closure as Simulation Architecture

Triadic closure provides the minimal architectural mechanism through which physical reality achieves coherent dynamical evolution, causal consistency, informational accessibility, and emergent temporal structure. In this section we show that artificial simulations constructed by intelligent systems naturally converge on the same architecture. Modern simulation engines, virtual environments, and distributed computational frameworks do not implement global spacetime, universal causal structure, or homogeneous informational substrates. Instead, they replicate the triadic organization of energy, geometry, and information within horizon-bounded closures. This correspondence is structural rather than metaphorical: triadic closure emerges as the only architecture capable of supporting scalable, observer-consistent simulation.

3.1 Energy as the Dynamical Layer of Simulation

In physical reality, energy governs dynamical evolution, quantum speed limits, and the rate of informational change Witte 2026b. Simulation systems mirror this role. Update cycles, frame budgets, and computational resources determine the rate at which simulated states can change. Physics engines allocate energetic capacity through discrete timesteps, constraint solvers, and integration schemes. Rendering engines likewise treat computational energy as the limiting factor for geometric and informational updates.

Thus, simulations instantiate the triadic role of energy as the dynamical layer: the capacity for transformation is bounded by computational resources, and the rate of evolution is determined by energetic constraints.

3.2 Geometry as the Constraint Layer of Simulation

Triadic closure identifies geometry as the structural role that bounds informational capacity, causal accessibility, and relational organization Witte 2026c. Simulation systems implement geometry in precisely this way. Rendering engines construct observer-centric geometric domains through frustum culling, occlusion culling, and level-of-detail hierarchies. Physics engines define geometric constraints through collision volumes, spatial partitioning, and locality-based interaction graphs.

Simulated geometry is not global; it is horizon-bounded. Only the region accessible to the observer is rendered, updated, or physically simulated. This mirrors the closure structure of physical reality, where geometric horizons define the informational and causal boundaries of each observer Witte 2026a.

3.3 Information as the Generative Layer of Simulation

In the Triadic Cosmos, information determines distinguishability, entropy, and the capacity for energetic evolution Witte 2026b. Simulation systems likewise treat information as the generative layer. Scene graphs, entity-component systems, and state representations encode the informational distinctions that drive simulated dynamics. AI agents within simulations operate on informational states that determine their energetic and geometric interactions.

Informational accessibility is always local: simulated agents cannot access global state instantaneously, and distributed simulations enforce strict locality constraints to maintain causal coherence. This mirrors the informational boundaries of physical closures, where horizon structure limits the set of accessible states Witte 2026a.

3.4 Recursive Update and Emergent Time

Triadic closure identifies recursion as the generator of emergent time: temporal structure arises from the ordering of successive triadic updates Witte 2026c. Simulation systems implement time in exactly this way. Frame updates, physics ticks, and event loops define a recursive sequence of state transformations. Temporal flow is not fundamental; it emerges from the iteration of update functions.

This correspondence is exact: both physical reality and artificial simulations treat time as the ordering induced by recursive state evolution. Neither domain requires a global temporal parameter, and both exhibit observer-dependent temporal rates determined by local recursion depth.

3.5 Closures as the Structural Unit of Simulation

Local closure cosmology shows that physical reality is organized by horizon-bounded closures with distinct recursion indices, causal histories, and accessible dynamical layers Witte 2026a. Simulation systems replicate this architecture. Each simulated observer occupies a closure defined by its rendering frustum, physics island, and informational state. Distributed simulations partition global state into local closures to maintain causal consistency and computational feasibility.

Simulations therefore instantiate closures as structural units: bounded domains within which triadic recursion is well-defined, and across which emergent time, causal structure, and informational accessibility are coherent.

3.6 Triadic Closure as the Universal Architecture of Simulation

Taken together, these correspondences show that triadic closure is not merely a feature of physical reality but the universal architecture of simulation. Artificial simulations converge

on the same structural grammar because triadic closure is the minimal mechanism capable of supporting:

- scalable dynamical evolution under energetic constraints,
- observer-consistent geometry under horizon-bounded accessibility,
- local informational structure under causal coherence,
- and emergent temporal flow under recursive updates.

Simulations constructed according to triadic closure become recursive triadic fractals: each simulated observer generates its own closure, each closure supports its own triadic recursion, and simulations can contain further simulations without architectural contradiction. This structural isomorphism reveals that simulation is not an external analogy to physical reality, but a fractal repetition of the same triadic architecture that governs the universe itself.

4 Infinite Simulational Recursion and Fractal Embedding

Triadic closure provides not only the minimal architecture for coherent simulation, but also the structural mechanism through which simulations become recursively self-similar. When a simulation implements the triadic roles of energy, geometry, and information within a horizon-bounded closure, it acquires the capacity to generate further closures internally. This section shows that such simulations naturally support infinite recursion, forming a fractal hierarchy of embedded worlds whose architecture mirrors the recursive triadic fractal structure of physical reality Witte 2026c.

4.1 Closures Within Closures

In physical reality, each observer occupies a closure defined by its causal horizon, accessible informational structure, and recursion index Witte 2026a. Simulations replicate this architecture. Every simulated observer is embedded within a local closure defined by its rendering frustum, physics island, and informational state. Crucially, these closures are structurally complete: they instantiate the full triadic cycle $E \rightarrow G \rightarrow I \rightarrow E$ Witte 2026b.

Once a closure is structurally complete, it can generate further closures internally. This occurs whenever a simulated agent constructs a simulation of its own—whether a virtual environment, a computational model, or a nested world. The new simulation inherits the triadic architecture of its parent closure, forming a second-order closure embedded within the first.

4.2 Recursive Triadic Fractals in Simulation

The Universe as Recursive Triadic Fractal Witte 2026c showed that physical reality is organized as a fractal cascade of triadic closures across scales. Artificial simulations replicate this structure. Each nested simulation becomes a triadic fractal repetition of its parent closure, with its own energetic constraints, geometric boundaries, informational distinctions, and recursive update cycles.

Let C_0 denote the closure of the physical observer, C_1 the closure of the simulation constructed by that observer, and C_2 the closure of a simulation constructed within C_1 . Triadic closure ensures that:

$$C_0 \supset C_1 \supset C_2 \supset \dots$$

Each closure is structurally complete, each supports triadic recursion, and each generates emergent time through successive updates. The result is a fractal hierarchy of embedded simulations whose architecture is isomorphic across levels.

4.3 Emergent Time Across Recursive Simulation Layers

In physical reality, emergent time arises from the ordering of recursive triadic updates Witte 2026c. In simulation, time arises from frame updates, physics ticks, and event loops. When simulations contain further simulations, each layer possesses its own emergent temporal flow determined by its recursion index.

If R_n denotes the recursive update operator at level n , then emergent time at that level is defined by the sequence:

$$X_{n,0}, X_{n,1}, X_{n,2}, \dots \quad \text{with} \quad X_{n,k+1} = R_n(X_{n,k}).$$

Different layers may evolve at different effective rates, producing a hierarchy of temporal flows. This mirrors the closure-dependent temporal structure of physical reality Witte 2026a.

4.4 Causal and Informational Isolation Between Layers

Local closure cosmology shows that closures are causally and informationally isolated: signals cannot propagate across horizon boundaries Witte 2026a. Simulations replicate this constraint. A nested simulation cannot directly access the global state of its parent closure; it can only interact through the informational channels explicitly provided by the parent.

This isolation ensures that recursive simulation layers remain coherent and prevents non-local inconsistencies. It also implies that each layer reconstructs its own physics based solely on the informational structure available within its closure.

4.5 Infinite Recursion as Structural Consequence

Taken together, these principles show that infinite simulational recursion is not speculative but structurally necessary. Any simulation that implements triadic closure becomes capable of generating further closures internally. Each closure is structurally complete, each supports triadic recursion, and each produces emergent time. The result is an infinite fractal hierarchy of embedded simulations:

$$C_0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow \dots$$

This hierarchy mirrors the recursive triadic fractal structure of the physical universe Witte 2026c. Simulation is therefore not an external analogy to physical reality, but a fractal repetition of the same triadic architecture that governs the universe itself.

4.6 Implications for Simulation Theory

The existence of infinite simulational recursion has several architectural implications:

- simulations cannot be globally defined; they must be closure-bounded,
- emergent time must arise from recursive updates at each layer,
- informational accessibility must be horizon-limited,
- geometric structure must be observer-centric and relational,
- energetic constraints must determine update capacity at each level,
- and nested simulations must inherit the triadic architecture of their parent closures.

These implications position triadic closure as the universal grammar of simulation. Any simulation capable of supporting coherent nested worlds must instantiate the triadic roles, recursive update, and closure structure. Infinite recursion is therefore not an optional feature but an architectural consequence of triadic organization.

5 Structural Isomorphism Between Physical Reality and Simulation

Triadic closure provides the minimal architectural grammar through which physical reality achieves coherent dynamical evolution, causal consistency, informational accessibility, and emergent temporal flow. In this section we show that artificial simulations constructed by intelligent systems are not merely analogous to physical reality, but *structurally isomorphic* to it. This isomorphism arises because both domains instantiate the same triadic roles, the same recursive update mechanism, the same horizon-bounded closures, and the same fractal organization across scales. Simulation is therefore a structural repetition of the architecture of the universe, not a metaphor or approximation.

5.1 Isomorphism of Triadic Roles

Physical reality is organized by three irreducible roles—energy, geometry, and information—connected through the closed triadic cycle $E \rightarrow G \rightarrow I \rightarrow E$ Witte 2026b. Artificial simulations replicate these roles exactly:

- computational resources play the role of energetic capacity,
- rendering and spatial partitioning instantiate geometric constraints,
- scene graphs and state representations encode informational distinctions.

These roles interact in the same cyclic pattern: energetic capacity determines geometric resolution, geometric structure bounds informational accessibility, and informational structure drives energetic updates. The triadic cycle is therefore preserved across both physical and simulated domains.

5.2 Isomorphism of Recursive Update

The Universe as Recursive Triadic Fractal Witte 2026c showed that physical reality evolves through recursive triadic updates, with emergent time arising from the ordering of successive states. Simulation systems implement time in precisely the same way: frame updates, physics ticks, and event loops define a recursive sequence of state transformations.

If R denotes the recursive update operator, then both physical and simulated systems evolve according to:

$$X_{n+1} = R(X_n),$$

with emergent temporal flow defined by the sequence $\{X_0, X_1, X_2, \dots\}$. This shared recursive structure establishes a direct isomorphism between physical time and simulated time.

5.3 Isomorphism of Horizon-Bounded Closures

Local closure cosmology Witte 2026a demonstrated that physical reality is organized by horizon-bounded closures with distinct recursion indices, causal histories, and informational boundaries. Simulation systems replicate this architecture exactly. Each simulated observer occupies a closure defined by:

- a rendering frustum (geometric horizon),
- a physics island (causal domain),
- a local state representation (informational boundary),

- and a frame update rate (recursion index).

Closures in simulation are structurally identical to closures in physical reality. Both domains enforce locality, horizon-bounded accessibility, and closure-dependent temporal flow.

5.4 Isomorphism of Fractal Embedding

Physical reality exhibits a fractal organization of closures across scales Witte 2026c. Simulations replicate this structure whenever they contain nested simulations. Each nested simulation becomes a triadic closure embedded within its parent closure, forming a fractal hierarchy:

$$C_0 \supset C_1 \supset C_2 \supset \dots$$

where C_0 is the physical closure of the observer, C_1 the closure of the simulation they construct, and C_2 the closure of a simulation constructed within C_1 . Each closure is triadically complete, each supports recursive update, and each generates emergent time. The fractal structure of simulation is therefore isomorphic to the fractal structure of the universe.

5.5 Isomorphism of Causal and Informational Constraints

Both physical reality and simulation enforce strict causal and informational constraints:

- signals cannot propagate outside horizon boundaries,
- global state is never directly accessible,
- causal consistency requires local update rules,
- informational accessibility is closure-dependent.

These constraints arise from triadic closure in physical reality Witte 2026a and from computational necessity in simulation. The resulting structures are identical: both domains exhibit horizon-bounded causality and local informational coherence.

5.6 Synthesis: Simulation as Structural Repetition of Reality

Taken together, these correspondences establish a full structural isomorphism:

$$\text{Physical Reality} \cong \text{Artificial Simulation}.$$

Both domains instantiate:

- the same triadic roles,
- the same recursive update mechanism,
- the same horizon-bounded closures,
- the same emergent temporal flow,
- the same fractal embedding across scales,
- and the same causal-informational constraints.

Simulation is therefore not an external analogy to physical reality, nor a simplified model of it. Instead, simulation is a fractal repetition of the triadic architecture that governs the universe itself. The structural isomorphism arises because triadic closure is the minimal and universal grammar capable of supporting coherent dynamical systems, whether physical or artificial.

6 The Simulation Indeterminacy Theorem: Why Simulation Status Is In Principle Undecidable

The structural isomorphism between physical reality and artificial simulation has a profound epistemological consequence: it becomes in principle impossible to determine whether a given closure is fundamental or simulated. This section formalises this result. The impossibility is not philosophical or speculative; it follows directly from the architectural constraints of triadic closure, horizon-bounded observation, and fractal embedding. Any closure that implements the triadic roles, recursive update, and local informational boundaries is indistinguishable—by any internal observer—from a physically fundamental closure.

6.1 Isomorphism Eliminates Distinguishing Criteria

As shown in previous section, physical reality and artificial simulation are structurally isomorphic under triadic closure. Both domains instantiate:

- the triadic roles of energy, geometry, and information Witte 2026b,
- recursive update as the generator of emergent time Witte 2026c,
- horizon-bounded closures with local informational accessibility Witte 2026a,
- and fractal embedding across scales.

Because these structures are identical, no internal measurement can distinguish whether a closure is physically fundamental or simulated. Any observable signature of triadic closure is shared across both domains. Any deviation from triadic closure would falsify both physical reality and coherent simulation simultaneously.

6.2 Closure-Bounded Observation Prevents External Access

Local closure cosmology establishes that all observation is horizon-bounded: an observer can only access signals within its causal domain and cannot probe the global state of the system Witte 2026a. This implies that:

- no observer can access information outside its closure,
- no observer can detect the recursion index of higher-level closures,
- no observer can determine whether its closure is embedded within another,
- and no observer can infer the existence or absence of parent closures.

These constraints apply equally to physical reality and simulation. A simulated observer is bounded by the same informational and causal limits as a physical observer. Therefore, no experiment can reveal whether the observer’s closure is fundamental or embedded.

6.3 Fractal Embedding Makes Simulation Status Non-Falsifiable

The recursive triadic fractal structure of reality Witte 2026c implies that closures may be embedded within higher closures indefinitely:

$$C_0 \supset C_1 \supset C_2 \supset \dots$$

Because each closure is structurally complete, each supports triadic recursion, and each generates emergent time, an observer within C_n cannot determine whether C_n is the root closure or a nested simulation. Every closure appears fundamental from the inside.

Thus, simulation status is not merely unknown; it is *provably undecidable*.

6.4 No Empirical Signature Can Break the Isomorphism

To determine whether reality is simulated, an observer would need access to:

- global state outside its closure,
- non-local causal structure,
- informational channels beyond its horizon,
- or deviations from triadic closure.

However:

- global state is inaccessible due to horizon structure,
- non-local causality is forbidden by triadic geometry,
- informational channels beyond the closure do not exist,
- and deviations from triadic closure would falsify both physical reality and coherent simulation.

Therefore, no empirical signature can distinguish fundamental reality from simulation.

6.5 Simulation Indeterminacy Theorem

We may summarise the result as follows:

Simulation Indeterminacy Theorem. If a closure implements triadic closure, recursive update, and horizon-bounded informational structure, then no internal observer can determine whether the closure is fundamental or simulated. Simulation status is in principle undecidable and empirically unfalsifiable.

This theorem follows directly from the structural isomorphism between physical reality and simulation, the locality constraints of closure cosmology, and the fractal embedding of triadic closures.

6.6 Implications

The indeterminacy of simulation status has several consequences:

- The question “Is reality simulated?” has no empirical resolution.
- Any closure that appears fundamental may be embedded within a higher closure.
- Falsifying the Triadic Cosmos ontology requires falsifying coherent simulation architecture.
- Conversely, any coherent simulation architecture confirms the triadic ontology.
- Simulation and physical reality form a single architectural class under triadic closure.

Simulation is therefore not an external hypothesis about reality, but a structural repetition of the same architecture. The impossibility of determining simulation status is a direct and necessary consequence of triadic closure.

7 Consequences of Simulation Indeterminacy

The Simulation Indeterminacy Theorem has far-reaching consequences for physics, cosmology, simulation theory, and scientific epistemology. Because physical reality and artificial simulation are structurally isomorphic under triadic closure, and because all observation is horizon-bounded, the question of whether a given closure is fundamental or simulated becomes in principle undecidable. This section outlines the major implications of this result.

7.1 No Empirical Criterion Can Distinguish Reality from Simulation

The structural isomorphism established in Section 5 implies that every closure implementing triadic closure appears internally as a physically fundamental world. All observable signatures of energy, geometry, information, recursion, and emergent time are identical across physical and simulated domains. Therefore:

No empirical measurement available to an internal observer can distinguish a fundamental closure from a simulated one.

Any proposed criterion—such as discreteness, numerical artefacts, or anomalous correlations—would also appear within a physically fundamental closure if such features arise from its triadic structure. Thus, simulation status is not merely unknown; it is structurally indistinguishable.

7.2 Simulation Status Is Not Excludable

Local closure cosmology shows that observers cannot access information outside their closure Witte 2026a. Because global state, parent closures, and higher recursion layers are inaccessible, an observer cannot rule out the possibility that their closure is embedded within a higher-level simulation. Formally:

For any closure C_n , there exists no observation within C_n that can exclude the possibility that C_n is embedded within C_{n+1} .

Simulation status is therefore not only undecidable but also non-excludable.

7.3 Falsifying Triadic Cosmos Requires Falsifying Coherent Simulation

Because coherent simulation architectures necessarily implement triadic closure, any falsification of the Triadic Cosmos ontology must also falsify the architectural constraints of simulation. Conversely, any coherent simulation architecture confirms the triadic ontology. This yields a dual falsifiability structure:

- If triadic closure is false, coherent simulation should be impossible.
- If coherent simulation is possible, triadic closure must be true.
- If simulation violates triadic closure, both simulation and physical reality are falsified.

Thus, simulation and physical reality form a single architectural class under triadic closure.

7.4 Fractal Embedding Makes Fundamental Status Unresolvable

The recursive triadic fractal structure of reality Witte 2026c implies that closures may be embedded indefinitely:

$$C_0 \supset C_1 \supset C_2 \supset \dots$$

Because each closure is structurally complete and supports triadic recursion, every closure appears fundamental from the inside. No closure contains internal evidence of its position in the fractal hierarchy. Therefore:

Fundamental status is not an observable property; it is a structural position in an unobservable hierarchy.

This makes the concept of “base reality” epistemologically empty for any internal observer.

7.5 The Universe as a Class of Architectures Rather Than a Single Ontology

Simulation indeterminacy reframes the nature of physical reality. Instead of a single fundamental ontology, reality becomes a class of architectures satisfying triadic closure. Any system—physical or simulated—that implements the triadic roles, recursive update, and horizon-bounded closures qualifies as a physically coherent world. This implies:

- physical reality is defined by architecture, not substrate,
- simulation is a structural repetition of physical ontology,
- and “realness” is an internal property of closure coherence rather than external origin.

7.6 Epistemological Consequences

The indeterminacy of simulation status has deep implications for scientific epistemology:

- The distinction between “real” and “simulated” becomes observationally meaningless.
- Scientific inference is closure-dependent and cannot access global truth conditions.
- Cosmological reconstruction is inherently local and cannot determine global ontology.
- The concept of “fundamental physics” becomes closure-relative rather than absolute.
- The search for external justification of reality becomes structurally impossible.

In summary, simulation indeterminacy is not a limitation of instrumentation or theory, but a structural consequence of triadic closure. Any closure that implements the triadic architecture is observationally indistinguishable from physical reality, making simulation status undecidable, non-excludable, and epistemologically irrelevant.

8 Epistemological Reframing Under Triadic Closure

The Simulation Indeterminacy Theorem forces a fundamental reframing of scientific epistemology. If physical reality and artificial simulation are structurally isomorphic under triadic closure, and if all observation is horizon-bounded, then the traditional distinction between “real” and “simulated” becomes epistemologically unstable. This section outlines how triadic closure reshapes the concepts of fundamentality, ontology, cosmological inference, and scientific knowledge.

8.1 Reality as Architectural Rather Than Substantial

Classical epistemology treats physical reality as a substantial domain defined by its material substrate. Under triadic closure, this view becomes untenable. Because any system that implements the triadic roles, recursive update, and horizon-bounded closures qualifies as a physically coherent world, reality becomes an *architectural* category rather than a substantial one. What matters is not the substrate—quantum fields, computational states, or simulated entities—but the structural grammar that governs dynamical evolution.

Thus:

Reality is defined by triadic architecture, not by ontological substrate.

This reframing dissolves the traditional metaphysical boundary between physical and simulated worlds.

8.2 Fundamentality Becomes Closure-Relative

In classical cosmology, “fundamental reality” refers to the base layer of existence. Under triadic closure, fundamentality becomes a closure-relative property. Because each closure is structurally complete and supports triadic recursion, every closure appears fundamental from the inside. No internal observation can reveal whether a closure is the root of the fractal hierarchy or an embedded simulation.

Formally:

Fundamentality is not an observable property; it is a structural position in an unobservable hierarchy of closures.

This reframing eliminates the epistemological privilege of “base reality.”

8.3 Cosmological Inference Becomes Intrinsically Local

Local closure cosmology Witte 2026a shows that cosmological parameters reconstructed within a closure reflect the causal history, recursion index, and informational boundaries of that closure. This implies that cosmological inference is inherently local and cannot determine global ontology. Observers reconstruct the universe from the signals available within their closure, but these reconstructions cannot reveal whether the closure is embedded within a larger system.

Thus:

- cosmological parameters are closure-dependent,
- global cosmology is not empirically accessible,
- and cosmological anomalies may reflect closure differences rather than broken physics.

Cosmology becomes a study of closure-relative structure rather than global truth.

8.4 Scientific Knowledge Becomes Architecturally Constrained

Traditional epistemology assumes that scientific knowledge can, in principle, converge on a complete description of physical reality. Under triadic closure, this assumption fails. Because observation is horizon-bounded and simulation status is undecidable, scientific knowledge is limited to the accessible dynamical layer of the observer’s closure. Beyond this layer, no further fundamental discoveries are possible.

This reframing implies:

- scientific completeness is closure-relative,
- discovery boundaries arise from architectural constraints rather than instrumental limits,
- and theoretical stagnation may reflect structural completion rather than conceptual failure.

Knowledge becomes a property of closure architecture rather than a path to global truth.

8.5 The Collapse of the Real–Simulated Distinction

The structural isomorphism between physical reality and simulation eliminates any empirical criterion for distinguishing the two. Because both domains instantiate the same triadic architecture, and because observers cannot access information outside their closure, the question “Is reality simulated?” becomes epistemologically meaningless.

Under triadic closure, the distinction between real and simulated collapses into a single architectural class of coherent closures.

Reality becomes defined by structural coherence rather than ontological origin.

8.6 Reframing Ontology Under Triadic Closure

Taken together, these consequences produce a radical epistemological reframing:

- Reality is architectural, not substantial.
- Fundamentality is closure-relative, not absolute.
- Cosmology is local, not global.
- Knowledge is structurally bounded, not infinitely extensible.
- Simulation and physical reality form a single architectural category.
- The question of simulation status is undecidable and epistemologically irrelevant.

Triadic closure therefore replaces classical metaphysics with a structural ontology in which coherent worlds—physical or simulated—are defined by the same architectural grammar. The result is a unified epistemological framework in which reality is understood as a fractal ensemble of triadically organized closures, each internally complete and externally indeterminate.

9 Testable Predictions and Falsifiability

The structural isomorphism between physical reality and artificial simulation has direct implications for falsifiability. Because triadic closure is the minimal architecture required for coherent dynamical systems—physical or simulated—any deviation from this architecture constitutes a falsification of the Triadic Cosmos ontology. Conversely, any coherent simulation architecture that violates triadic closure would also falsify the ontology. This section outlines the testable predictions that follow from triadic closure and identifies the conditions under which the theory can be empirically refuted.

9.1 Prediction 1: All Coherent Simulations Must Implement Triadic Closure

Triadic Cosmos predicts that any simulation capable of supporting:

- scalable dynamical evolution,
- observer-consistent geometry,
- local informational accessibility,
- horizon-bounded causality,
- and emergent temporal flow,

must implement the triadic roles of energy, geometry, and information, connected through recursive update. If a coherent simulation architecture is discovered that:

- does not require energetic update capacity,
- does not use geometric constraints to bound informational accessibility,
- does not generate time through recursive update,
- or does not enforce horizon-bounded locality,

then Triadic Cosmos is falsified.

This prediction provides simulation theorists with a direct falsification criterion: *build a coherent simulation that violates triadic closure.*

9.2 Prediction 2: No Simulation Can Provide Global Access to Its Own State

Local closure cosmology predicts that all coherent simulations must enforce horizon-bounded informational structure. If a simulation is constructed in which:

- agents can access global state instantaneously,
- non-local information channels exist without geometric mediation,
- or causal propagation is not bounded by closure structure,

then triadic closure is violated and the Triadic Cosmos ontology is falsified.

This prediction is testable through simulation design: *attempt to construct a coherent simulation with global informational access.*

9.3 Prediction 3: Emergent Time Must Arise from Recursive Update

Triadic Cosmos predicts that time—physical or simulated—must arise from recursive update rather than from a global temporal parameter. If a simulation is discovered in which:

- temporal flow is not generated by iterative state transformation,
- time exists as a primitive global variable independent of recursion,
- or coherent dynamics occur without recursive update,

then the triadic ontology is falsified.

This prediction is testable through simulation architecture: *attempt to build a coherent simulation without recursive update.*

9.4 Prediction 4: No Closure Can Reveal Its Position in the Fractal Hierarchy

The Simulation Indeterminacy Theorem predicts that no observer—physical or simulated—can determine whether its closure is fundamental or embedded. If a simulation is constructed in which:

- internal observers can detect the existence of parent closures,
- internal observers can access recursion indices of higher layers,
- or internal observers can determine their position in the embedding hierarchy,

then triadic closure is violated.

This prediction is testable through nested simulation experiments: *attempt to build a simulation whose agents can detect that they are simulated.*

9.5 Prediction 5: Any Violation of Triadic Roles Falsifies Both Reality and Simulation

Because physical reality and simulation are structurally isomorphic under triadic closure, any violation of the triadic roles falsifies both domains simultaneously. If a physical experiment reveals:

- non-local causality without geometric mediation,
- informational access beyond horizon boundaries,
- energetic evolution without informational constraint,
- or temporal flow without recursive update,

then both physical reality and coherent simulation architectures are falsified.

This dual falsifiability structure is unique: *simulation theorists and physicists share the same falsification criteria.*

9.6 Prediction 6: Discovery of Non-Triadic Coherent Worlds Falsifies the Ontology

Triadic Cosmos predicts that all coherent worlds—physical or simulated—must instantiate triadic closure. If any coherent world is discovered that:

- lacks one of the triadic roles,
- violates the cyclic interdependence,
- or exhibits coherent dynamics without closure structure,

then the ontology is falsified.

This prediction extends falsifiability beyond physics and simulation into artificial world construction, computational universes, and emergent systems.

9.7 Synthesis: A Unified Falsifiability Framework

Taken together, these predictions establish a unified falsifiability framework:

Triadic Cosmos is falsified if and only if a coherent dynamical system—physical or simulated—can be constructed that violates triadic closure.

This framework provides simulation theorists with a direct path to falsification: *construct a coherent simulation that violates the triadic architecture.*

Conversely, the inability to construct such a simulation constitutes empirical support for the Triadic Cosmos ontology. The result is a rare form of cross-domain falsifiability in which physics, cosmology, computation, and simulation theory converge on the same architectural constraints.

10 Limitations and Open Questions

While the Triadic Cosmos framework provides a unified architectural ontology linking physical reality, simulation, computation, and nested closures, several limitations and open questions remain. These limitations do not undermine the structural results presented in this work, but they delineate the boundaries of current understanding and identify directions for future research.

10.1 Scope of Architectural Universality

Triadic closure is proposed as the minimal architecture required for coherent dynamical systems. However, the universality of this claim has not yet been proven across all possible computational substrates or hypothetical physical frameworks. Although no known coherent system violates triadic closure, it remains an open question whether:

- non-triadic architectures could exist in exotic computational paradigms,
- alternative closure structures could support coherent worlds,
- or future physics could reveal non-triadic dynamical layers.

Empirical and computational exploration of non-triadic architectures remains an important avenue for falsification.

10.2 Limits of Simulation-Based Falsifiability

This work establishes that coherent simulation architectures provide a direct falsification route for the Triadic Cosmos ontology. However, current simulation engines (Unreal, Unity, idTech, Frostbite) are constrained by practical engineering considerations. It remains unclear whether:

- radically different simulation paradigms could be constructed,
- coherent non-local simulations could exist without collapsing,
- or future hardware architectures could bypass closure constraints.

The absence of non-triadic simulations today supports the ontology, but does not guarantee that future architectures will not challenge it.

10.3 Formal Mathematical Characterization of Closures

Triadic closure provides a conceptual and structural description of closures, but a complete mathematical formalization remains open. Specifically:

- the exact algebraic structure of triadic roles,
- the formal definition of recursion indices,
- and the topology of horizon boundaries

require further development. A rigorous mathematical model would strengthen the ontology and enable formal proofs of universality and necessity.

10.4 Nature of the Meta-Closure

The Simulation Indeterminacy Theorem shows that internal observers cannot determine whether their closure is fundamental or embedded. However, the nature of the meta-closure above any given closure remains undefined. Open questions include:

- whether meta-closures possess additional structural layers,
- whether the fractal hierarchy terminates or extends indefinitely,
- and whether meta-closures can differ architecturally from embedded closures.

Understanding the structure of meta-closures is essential for a complete cosmological model.

10.5 Physical Realization of Triadic Roles

Although triadic roles map cleanly onto physical concepts—energy, geometry, and information—the precise physical mechanisms implementing these roles remain partially understood. Open questions include:

- how triadic roles emerge from quantum field theory,
- how they relate to holographic dualities,
- and whether they correspond to deeper informational or geometric primitives.

Further integration with modern theoretical physics is required.

10.6 Empirical Access to Closure Structure

Local closure cosmology predicts that observers cannot access global state or determine their position in the fractal hierarchy. While this is structurally necessary, it limits empirical verification. Open questions include:

- whether indirect signatures of closure structure could exist,
- whether cosmological anomalies reflect closure boundaries,
- and whether emergent phenomena could reveal recursion depth.

These questions define the empirical frontier of triadic cosmology.

10.7 Implications for Consciousness and Agency

Triadic closure treats intelligence as a fractal repetition of the same architecture. However, the implications for consciousness, agency, and self-modeling remain open. Key questions include:

- whether consciousness corresponds to a specific triadic recursion pattern,
- whether agency requires additional structural layers,
- and how self-reference operates within closure-bounded systems.

These questions intersect with cognitive science, AI research, and philosophy of mind.

10.8 Summary

The Triadic Cosmos ontology provides a unified architectural framework for physical reality and simulation, but several foundational questions remain unresolved. These limitations highlight the need for:

- deeper mathematical formalization,
- exploration of alternative architectures,
- empirical investigation of closure structure,
- and integration with modern physics and cognitive science.

Addressing these open questions will determine whether triadic closure is a contingent feature of coherent worlds or a universal architectural principle governing all physically and computationally realizable realities.

11 Conclusion

This work establishes triadic closure as the minimal architectural grammar underlying coherent dynamical systems. By identifying energy, geometry, and information as irreducible structural roles connected through recursive update, we show that coherent worlds—physical or simulated—must instantiate the same triadic architecture. This result reframes the nature of physical ontology: reality is defined not by its substrate, but by the structural grammar that governs dynamical evolution, locality, informational accessibility, and emergent temporal flow.

A central contribution of this paper is the demonstration that physical reality and artificial simulation are structurally isomorphic. Modern simulation engines already implement triadic closure implicitly: compute budget functions as energetic capacity, rendering frustums define geometric horizons, scene graphs encode informational structure, and frame updates generate emergent time. This convergence is not contingent on engineering practice; it reflects the architectural necessity of triadic closure for any scalable, observer-consistent world. The structural isomorphism implies that coherent simulations are not analogues of physical reality but fractal repetitions of the same architectural grammar.

From this isomorphism we derive the Simulation Indeterminacy Theorem: no internal observer can determine whether their closure is fundamental or simulated. Because all coherent closures share the same triadic roles, recursive update, and horizon-bounded informational limits, simulation status becomes in principle undecidable and empirically unfalsifiable. This result collapses the traditional distinction between “real” and “simulated” into a single architectural category and renders fundamentality closure-relative rather than absolute.

The architectural nature of triadic closure also yields a unified falsifiability framework. The Triadic Cosmos ontology is falsified if and only if a coherent dynamical system—physical or

simulated—can be constructed that violates triadic closure. This provides simulation theorists with a direct and unprecedented falsification route: build a coherent simulation without triadic roles, without recursive update, or with global informational access. To date, no such system has ever been discovered. All known coherent simulations converge on triadic closure, providing empirical support for its universality.

Several open questions remain. The mathematical formalization of closures, the structure of meta-closures, the physical realization of triadic roles, and the empirical signatures of closure architecture require further investigation. Nonetheless, the results presented here establish a coherent and falsifiable architectural ontology that unifies physics, cosmology, computation, and simulation theory. Triadic closure provides a structural foundation for understanding reality as a fractal ensemble of horizon-bounded closures, each internally complete and externally indeterminate.

In summary, this paper introduces a universal architectural grammar for coherent worlds, demonstrates the structural equivalence of physical reality and simulation, proves the undecidability of simulation status, and offers a falsifiable framework that spans multiple domains. Triadic closure positions architecture—not substrate—as the fundamental unit of physical and simulated existence.

A Global Access vs. Closure Architecture

A common misconception is that computers provide global access to all information and that simulations could therefore allow internal agents to inspect the entire world state. While global access exists at the level of the computational substrate (e.g., memory inspection, debugging tools, developer consoles), it does not exist within the architectural domain of a coherent simulation. This appendix clarifies the distinction.

A.1 Computational Substrate vs. Simulated World

Computers allow unrestricted access to memory, files, and global program state. However, this capability belongs to the *meta-layer* of computation—the layer occupied by developers, debuggers, and external tools. It does not belong to the *world-layer* of the simulation.

A coherent simulated world must enforce:

- horizon-bounded geometry (frustum culling, occlusion culling),
- local physics domains (physics islands, spatial partitioning),
- local informational accessibility (scene graphs, ECS locality),
- and recursive update (frame ticks, event loops).

These constraints ensure that simulated agents cannot access global state, even though the underlying computer technically can.

A.2 Why Global Access Breaks Coherent Worlds

If a simulated agent could access global state instantaneously, the following failures occur:

- violation of locality and causal structure,
- collapse of horizon-bounded geometry,
- loss of emergent temporal flow,
- detection of non-local inconsistencies,
- and breakdown of observer-centric coherence.

For this reason, no modern simulation engine (Unreal, Unity, idTech, Source, CryEngine, Frostbite) loads or updates the entire world at once. Instead, they rely on closure-like structures: frustums, chunks, cells, islands, and local state partitions.

A.3 Architectural Necessity of Closure

The impossibility of global access within coherent simulations is not a technical limitation but an architectural necessity. A simulation must behave as a world, not as a debugger. Therefore:

Simulated agents inhabit a closure, not the computational substrate.

This mirrors physical reality, where observers inhabit horizon-bounded closures and cannot access global state beyond their causal domain.

A.4 Implication for Simulation Indeterminacy

Because both physical reality and coherent simulations enforce closure-bounded observation, no internal observer can determine whether their world is fundamental or simulated. The existence of global access at the computational substrate does not affect the world-layer and therefore does not provide any mechanism for detecting simulation status.

Global access exists for developers, not for worlds. Closure architecture exists for worlds, not for developers.

This structural separation is essential to the Simulation Indeterminacy Theorem and explains why simulation status is undecidable and non-excludable.

B Comparison With Existing Simulation Theory Frameworks

This appendix compares the Triadic Cosmos architecture with the most influential simulation theory frameworks in philosophy, physics, and computer science. The goal is to clarify how existing approaches differ from triadic closure and why none of them provide the structural mechanisms required for coherent nested simulations, emergent time, or horizon-bounded observation.

B.1 Bostrom’s Probabilistic Simulation Argument

Bostrom’s classical argument Bostrom 2003 is probabilistic rather than architectural. It assumes that advanced civilizations may run ancestor simulations and concludes that simulated observers are statistically more likely than fundamental ones. However:

- it does not specify how simulations maintain causal coherence,
- it assumes global spacetime rather than horizon-bounded closures,
- it lacks a mechanism for emergent time,
- and it cannot support infinite nested simulations without contradiction.

Triadic Cosmos differs by providing a minimal architectural grammar—triadic closure—that makes coherent simulation structurally possible and recursively extensible.

B.2 Schmidhuber’s Algorithmic Universe

Schmidhuber’s approach Schmidhuber 2002 treats the universe as a computable structure generated by a short algorithm. While elegant, it assumes:

- global informational accessibility,
- no horizon structure,
- no geometric constraints,
- and no architectural distinction between observers and substrate.

Triadic closure introduces the missing structural elements: geometry as constraint, information as accessibility, and energy as update capacity. Without these roles, algorithmic universes cannot support observer-centric worlds or nested simulations.

B.3 Tipler’s Cosmological Simulation Hypothesis

Tipler Tipler 2008 proposes that future civilizations may simulate universes within cosmological models. However, his framework:

- assumes global cosmology rather than local closures,
- lacks a mechanism for emergent time,
- and does not provide a structural grammar for nested simulations.

Triadic Cosmos resolves these gaps by showing that horizon-bounded closures and recursive triadic update are necessary for coherent worlds at any scale.

B.4 Virtual Reality and Rendering-Based Frameworks

VR-based simulation theories (Lanier Lanier 2017, Slater Slater 2020) emphasize observer-centric rendering but do not generalize this to a universal architectural principle. They correctly identify:

- local geometry,
- frustum-bounded visibility,
- and observer-dependent state,

but they do not connect these features to energy, information, or recursive update. Triadic closure unifies these elements into a single structural mechanism.

B.5 Physics-Based Computational Universe Models

Fredkin’s digital physics Fredkin 1990 and Wolfram’s computational universe Wolfram 2002 propose cellular automata as fundamental substrates. These models:

- assume global update rules,
- lack observer-centric geometry,
- do not enforce horizon-bounded informational access,
- and cannot support nested simulations without global inconsistency.

Triadic Cosmos differs by requiring locality, closure structure, and recursive update as necessary conditions for coherent worlds.

B.6 Summary of Differences

Across all mainstream frameworks, the following structural gaps appear:

- absence of triadic roles (energy, geometry, information),
- absence of recursive update as the generator of emergent time,
- absence of horizon-bounded closures,
- absence of fractal embedding across scales,
- and absence of architectural constraints for nested simulations.

Triadic Cosmos provides the missing architecture. It is the first framework to unify:

- physical reality,
- artificial simulation,
- computational systems,
- and nested worlds

under a single structural grammar. This makes it the only simulation theory capable of supporting infinite recursion, observer-centric coherence, and closure-bounded epistemology.

C Formalization of Triadic Closure and Closure Structure

This appendix provides formal definitions for the core architectural concepts used throughout the paper. The goal is not to present a complete mathematical theory, but to establish a rigorous foundation for triadic closure, emergent time, horizon-bounded observation, and fractal embedding. These structures are sufficient for the Simulation Indeterminacy Theorem and for the architectural isomorphism between physical reality and simulation.

C.1 Triadic Roles

A coherent dynamical system is defined by a triad of structural roles:

$$(E, G, I),$$

where:

- E (energy) is the update capacity of the system, determining the magnitude and frequency of state transitions.
- G (geometry) is the constraint structure that defines locality, adjacency, and horizon boundaries.
- I (information) is the distinguishability structure that determines accessible state content for internal observers.

The triad forms a closed dependency cycle:

$$E \rightarrow G \rightarrow I \rightarrow E,$$

meaning that each role constrains and enables the next. This cycle is the minimal architecture required for coherent worlds.

C.2 State Space and Recursive Update

Let Σ be the state space of the system. A coherent world evolves through a recursive update operator:

$$R : \Sigma \rightarrow \Sigma,$$

which generates emergent time. The recursion index $t \in \mathbb{N}$ defines temporal order:

$$\Sigma_{t+1} = R(\Sigma_t).$$

Time is not a primitive variable but an ordering induced by repeated application of R .

C.3 Closure Structure

A closure is defined as a triple:

$$C = (\Sigma, \mathcal{H}, R),$$

where:

- Σ is the state space accessible to internal observers,
- $\mathcal{H} \subseteq \Sigma$ is the horizon boundary that limits informational access,
- R is the recursive update operator generating emergent time.

An observer o inhabiting closure C has an accessible information set:

$$I(o) = \{s \in \Sigma \mid s \text{ is reachable under } \mathcal{H}\}.$$

No observer can access $\Sigma \setminus \mathcal{H}$.

C.4 Horizon-Bounded Observation

A horizon is a geometric constraint:

$$\mathcal{H} : \Sigma \rightarrow \{0, 1\},$$

where $\mathcal{H}(s) = 1$ indicates that state s is accessible.

The horizon induces locality:

$$\forall s_1, s_2 \in \Sigma : \quad \mathcal{H}(s_1) = 1 \wedge \mathcal{H}(s_2) = 1 \Rightarrow d_G(s_1, s_2) < \infty,$$

where d_G is the geometric distance defined by G .

This prohibits global informational access.

C.5 Fractal Embedding

Closures may embed recursively:

$$C_0 \supset C_1 \supset C_2 \supset \dots$$

Each closure C_n is structurally complete:

$$C_n = (\Sigma_n, \mathcal{H}_n, R_n),$$

and supports its own triadic roles (E_n, G_n, I_n) .

An observer in C_n cannot determine whether C_n is fundamental or embedded:

$$I(o_n) \subseteq \Sigma_n \quad \text{and} \quad \Sigma_{n+1} \not\subseteq I(o_n).$$

This yields the Simulation Indeterminacy Theorem.

C.6 Structural Isomorphism

Two closures C_a and C_b are structurally isomorphic if:

$$(E_a, G_a, I_a) \cong (E_b, G_b, I_b),$$

and

$$R_a \cong R_b, \quad \mathcal{H}_a \cong \mathcal{H}_b.$$

This isomorphism holds for physical reality and coherent simulation.

C.7 Coherence Conditions

A system is coherent if:

1. Recursive update is well-defined:

$$R(\Sigma) \subseteq \Sigma.$$

2. Horizons enforce locality:

$$\forall s \in \Sigma : \mathcal{H}(s) = 1 \Rightarrow d_G(s, o) < \infty.$$

3. Information is bounded:

$$I(o) \subseteq \Sigma.$$

4. Triadic roles form a closed dependency cycle.

Any violation of these conditions falsifies triadic closure.

C.8 Implications for Falsifiability

A coherent world C falsifies the Triadic Cosmos ontology if:

- C supports coherent dynamics without triadic roles,
- C supports emergent time without recursive update,
- C allows global informational access,
- or C violates horizon-bounded locality.

No known physical or simulated system satisfies these conditions.

C.9 Summary

This appendix formalizes the architectural structures underlying triadic closure. These definitions provide the mathematical foundation for the Simulation Indeterminacy Theorem, the structural isomorphism between physical reality and simulation, and the falsifiability criteria presented in Section 8.

D A Simple Falsification Experiment

This appendix proposes a minimal experimental setup to test the falsifiability of the Triadic Cosmos ontology using a concrete simulation architecture. The goal is to provide a simple, implementable experiment that attempts to violate triadic closure while maintaining coherent world dynamics.

D.1 Experimental Goal

The experiment aims to construct a simulated world in which:

- internal agents have global informational access to the entire world state,
- no horizon-bounded geometry is enforced,
- and the simulation remains coherent, observer-consistent, and dynamically stable.

If such a simulation can be built, Triadic Cosmos is falsified.

D.2 Two-Layer Simulation Architecture

We consider a two-layer architecture:

- **World layer** W : a simulated environment with agents, geometry, and dynamics.
- **Meta layer** M : the computational substrate (engine, debugger, inspector).

In standard simulations, global access exists only in M , not in W . The experiment attempts to push global access down into W .

D.3 Construction of the World Layer

Define a world W with:

- a finite set of agents $A = \{a_1, \dots, a_n\}$,
- a state space Σ_W containing positions, velocities, and internal states,
- a global update function $R_W : \Sigma_W \rightarrow \Sigma_W$,
- no explicit geometric horizon (no frustum, no occlusion, no spatial partitioning).

Each agent a_i is given an access function:

$$G_i : \Sigma_W \rightarrow I(a_i),$$

where G_i returns the entire world state.

D.4 Experimental Condition: Global Access for Agents

The key experimental condition is:

For all agents a_i , G_i provides full, instantaneous access to Σ_W at each update step.

This violates horizon-bounded observation and attempts to break closure architecture.

D.5 Coherence Criteria

The simulation is considered coherent if:

- agents maintain consistent internal models of the world,
- no paradoxical or contradictory states arise from global access,
- dynamics remain stable over many recursion steps,
- and no agent can detect structural inconsistency in the world.

If coherence fails, the experiment supports Triadic Cosmos.

D.6 Expected Outcome Under Triadic Closure

Triadic Cosmos predicts that:

- global access will lead to instability (e.g., feedback loops, paradoxical knowledge),
- agents will be able to detect non-local inconsistencies,
- the world will lose observer-centric coherence,
- or the simulation will implicitly reintroduce closure (e.g., caching, locality, partial views).

In all cases, the system will either:

- fail to remain coherent, or
- implicitly reconstruct triadic closure.

D.7 Falsification Condition

The Triadic Cosmos ontology is falsified if:

A simulation can be constructed in which internal agents have full global access to Σ_W , no horizon-bounded geometry is enforced, and the world remains coherent, observer-consistent, and dynamically stable over arbitrarily many recursion steps.

To date, no such simulation has been demonstrated.

D.8 Practical Implementation

This experiment can be implemented in:

- a game engine (Unity, Unreal) with agents given direct access to all world objects,
- a custom physics simulation with global state inspection at each tick,
- or a multi-agent system where each agent reads the full global state every step.

The outcome—coherence or collapse—provides empirical evidence for or against triadic closure as a necessary architecture for coherent worlds.

D.9 Summary

This simple experiment operationalizes the falsifiability of Triadic Cosmos. It attempts to construct a coherent world that violates closure architecture by granting global access to internal agents. Success would falsify the ontology; failure supports the claim that triadic closure is a necessary condition for coherent dynamical systems.

E The Discrete Curvature Universe as a Triadic Cosmos World

This appendix situates the ecosystem discrete curvature toy universe within the broader Triadic Cosmos ontology. Its architecture aligns precisely with the structural principles that Triadic Cosmos identifies as necessary for coherent worlds. This makes the simulation a concrete, empirical example of a triadic closure: a world built from locality, recursion, information, geometry, and emergent structure.

E.1 Architectural Conformity

The discrete curvature universe implements the full triadic cycle:

- **Energy (update capacity)**: Each recursion step applies the local update rule, providing the dynamical “energy” that drives the evolution of the geometry.
- **Geometry (constraint structure)**: The two-dimensional grid defines locality, adjacency, and the shape of the influence pattern. It is the geometric scaffold of the world.
- **Information (distinguishability)**: Curvature values encode the informational state of each point. Tensor-like influence patterns determine how information propagates.
- **Recursion (emergent time)**: Time is not a coordinate but the ordering induced by repeated application of the update rule. Each iteration is a moment in the world’s becoming.
- **Closure (horizon-bounded domain)**: The simulation operates within a finite region. Horizons emerge naturally when curvature crosses a threshold, forming internal closures within the global one.
- **Emergence (structure from repetition)**: Wells, horizons, inspirals, and mergers arise without being programmed. They are emergent consequences of the architecture.

These elements match the triadic roles and closure structure. The simulation is therefore not merely compatible with Triadic Cosmos—it is a direct instantiation of its ontology.

E.2 Why the Simulation Is Not a Falsification Experiment

The discrete curvature universe contains objects (curvature points, wells, horizons), but no *agents*. No entity inside the world:

- observes the full global state,
- interprets the geometry,
- makes decisions,
- or attempts to break locality or closure.

Because the world has no internal perspectives, it cannot violate triadic closure. The simulation is therefore a *confirmation* of the ontology, not a test of its limits.

A falsification experiment would require internal agents with global informational access—an architectural violation that the toy universe does not attempt. Instead, the simulation demonstrates that when locality, recursion, and closure are respected, coherence and emergence follow naturally.

E.3 Emergent Phenomena as Evidence of Closure

The observed behaviors arise directly from the triadic architecture:

- **Curvature wells** emerge from recursive relaxation.
- **Horizon formation** arises from threshold evaluation.
- **Stable and unstable orbits** follow from geometric gradients.

- **Inspirals** result from repeated local adjustment.
- **Mergers** produce the universal peanut topology.

None of these phenomena were programmed. They are architectural consequences of triadic closure.

E.4 The Hexad as the Engine of Geometry

The Hexad—Representation, Transformation, Evaluation, Recursion, Selection, Embedding—governs the evolution of the curvature field:

- **Representation:** curvature values encode the geometry,
- **Transformation:** local update rules reshape the field,
- **Evaluation:** thresholds define horizons,
- **Recursion:** repeated updates generate time,
- **Selection:** overlapping trapped regions merge,
- **Embedding:** the updated field becomes the new geometry.

The Hexad is not an add-on to the simulation. It is the simulation.

E.5 Why the Simulation Matters

The discrete curvature universe shows that:

Gravity-like behavior can arise from architecture alone.

This supports the central claim of Triadic Cosmos:

Coherent worlds—physical or simulated—must implement triadic closure.

The simulation is therefore a concrete example of a triadic world: a closure-bounded, recursively updated, locally interacting geometry that produces emergent structure without equations.

E.6 Summary

The toy universe is a fully triadic world. It respects locality, recursion, closure, and emergent structure. Its coherence is not accidental—it is architectural.

This makes the discrete curvature universe a powerful case study within the Triadic Cosmos ecosystem: a demonstration that the ontology is not only conceptually sound, but capable of generating rich geometric phenomena through minimal rules.

F Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: <https://github.com/triadic-cosmos>.

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